

COMS 4776 Lecture 13: Transformer Variants & State-Space Models

Richard Zemel

Current Transformer Research

- Transformer variants
 - Efficiency gains: Linear transformers, Mixture of Experts
 - Memory gains: state-space models
- Transformers & interpretability
- Adapting pre-trained transformers:
 - Tuning mechanisms for open and closed models
 - Tuning objectives: control, personalization
- Model uncertainty

Linear Transformers

- The standard linear transformer (Katharopoulos et al, 2020) formulation uses the kernel trick to linearize the attention computation:

$$\text{LinearAttention}(Q, K, V) = \phi(Q)(\phi(K)^T V)$$

- where $\phi()$ is a feature map/kernel function applied element-wise, and Q, K, V are the query, key, and value matrices
- The key insight is changing the order of operations to compute $(K^T V)$ first, followed by multiplication with Q , reducing complexity from $O(n^2)$ to $O(n)$.
 - $\phi(Q)$ is $n \times d$, and $(\phi(K)^T V)$ is $d \times d$
 - So their product is $n \times d$, which requires $O(nd)$ operations

Common Kernel Functions

Several kernel functions $\phi()$ are commonly used:

- **ReLU Kernel:** $\phi(x) = \text{ReLU}(x)$
Simple and efficient, used in the original Linear Transformer paper
- **ELU Kernel:** $\phi(x) = \text{ELU}(x) + 1$
For negative x , $\text{ELU}(x) = \alpha(\exp(x) - 1)$
Used in Transformers are RNNs paper for better performance
- **Softmax Kernel Approximation:** $\phi(x) = \exp(x) / \sqrt{(d_k)}$
Approximates standard softmax attention
- **Random Feature Maps:** $\phi(x) = \exp(-||x||^2/2)$

Advantages of Linear Transformers vs. Scaled Dot-Product Attention

- Computational Efficiency: $O(n)$ time & space complexity, compared to $O(n^2)$ for SDPA $\rightarrow$ significantly more efficient for long sequences, large amounts of data
- Expensive to train and run inference with, especially as sequence length increases.
- Recurrent Formulation: Many linear attention variants can be computed recurrently, enabling efficient autoregressive generation and streaming applications.

Advantages of Scaled Dot-Product Attention

- Expressiveness: Can capture complex dependencies and relationships between any pair of tokens in the sequence.
- Strong Empirical Performance: Forms the backbone of most state-of-the-art transformer models.
- Well-Optimized: Highly optimized implementations on modern hardware (GPUs/TPUs).

Other Transformer Variants

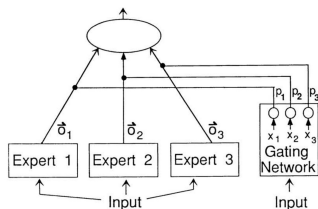
- Sparse attention: instead of attending to all tokens in sequence, each token attends only to a small subset of other tokens, e.g., Longformer (Beltagy et al, 2020) uses a sliding window approach.
- Approximate full attention matrix by low-rank matrix, e.g., Linformer (Wang et al, 2020) projects attention matrix into lower-dimensional space

Mixture of Experts & Transformers

- Main idea: build structure into MLP component of transformer block, to speed up pre-training and inference
- Example: Mixtral 8x7B from Dec 2023
- Structure is based on old neural network idea: mixture of experts

Mixture of Experts

- Papers from 1990s: Jacobs, Jordan, Nowlan, Hinton



- error: $-\log$ prob of generating desired output under a mixture of gaussians model of distribution of outputs given current input
- output of each expert specifies a mean
- outputs of the gating network specify the experts' mixing proportions
- both functions of the current input

$$E^c = \sum_k p_k^c \exp(-||\mathbf{t}^c - \mathbf{o}_k^c||^2 / 2\sigma^2)$$

Mixture of Experts: Learning

- gradient descent in E has two effects:
 - ① raises mixing proportion of experts that do better than average in predicting the desired output
 - ② makes each expert better at predicting targets for cases for which it has high mixing proportion
- after learning, gating network typically assigns mixing proportion near 1 to one expert on each case
- towards the end of learning, each expert can focus on modelling cases it is good at

Mixture of Experts & Transformers

Wins from MOEs?

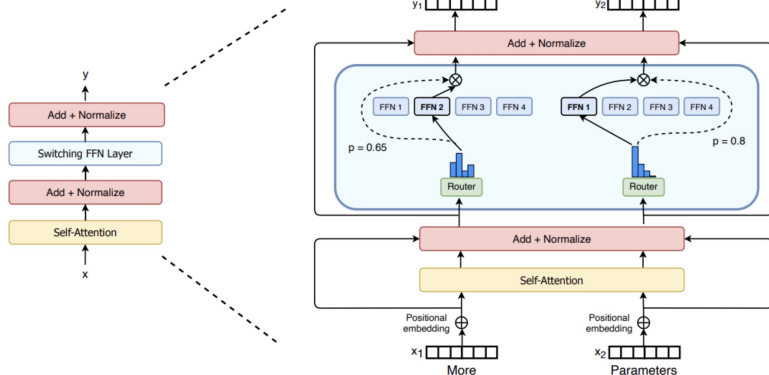
Mixture of Experts & Transformers

Wins from MOEs?

- Sparsity: not all tokens need to be processed in all network modules
- Choose only a few experts per input: top- k in gating network output
- Interpretability: specialization of network components
- Parallelism

Switch Transformers

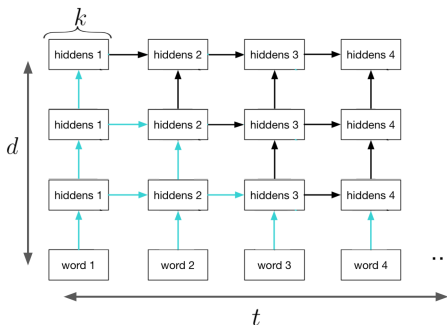
- Simplify the top- k gating output to top-1
- Multiple heads



State-Space Models

Transformer alternative for tasks involving time-series, sequential data

- Fundamental idea: Latent states evolve over time according to some dynamics, and observations are noisy measurements of these states



State-Space Models

Equation for a basic state-space model:

- State Transition: $\mathbf{x}_t = A\mathbf{x}_{t-1} + B\mathbf{u}_t + \mathbf{w}_t$
- Observation: $\mathbf{y}_t = C\mathbf{x}_t + \mathbf{v}_t$

Here:

- $\mathbf{x}_t$ is the latent state at time t
- $\mathbf{y}_t$ is the observed output
- A governs how the state evolves
- B represents how control inputs influence the state
- $\mathbf{w}_t$ and $\mathbf{v}_t$ are noise terms

SSMs often require fewer parameters, can operate in continuous settings, and are flexible in modeling underlying generative process

Mamba

SSM architecture that combines efficiency of SSMs with expressiveness of attention-based transformers (Gu & Dao, 2023)

- *Data-Dependent Parameters*: Computes state transition parameters (A, B, C matrices) as functions of input data
- Provides some adaptive qualities of attention without quadratic cost
- RNNs must compress all information into a fixed-size state vector
- Mamba determines what information to keep based on importance to task, can maintain relevant context over longer sequences

$$\begin{aligned}h(t) &= A(x(t))h(t-1) + B(x(t))x(t) \\ y(t) &= C(X(t))h(t)\end{aligned}$$

Mamba Beats Transformers

- Long sequence modeling: Mamba handles very long sequences more efficiently due to linear scaling versus Transformer's quadratic scaling.
- Streaming tasks: Tasks requiring continuous data processing benefit from Mamba's state-space model architecture.
- Time-series analysis: Mamba shows strong performance on sequential data like time-series forecasting.

Transformers Beat Mamba

- Flexible attention: Transformers excel at tasks requiring understanding of relationships between elements regardless of position
- Parallelization: Transformer attention can be highly parallelized during training
- Multi-modal tasks: Transformer models have proven effective for tasks involving multiple types of data (text, images, audio)
- Transfer learning: Transformers has demonstrated strong transfer learning capabilities across domains
- Short-to-medium sequence tasks: For sequences that aren't very long, Transformers often produce better results
- Maturity of implementation: Transformers have had years of optimization across frameworks and hardware